SECTION A

MULTIPLE CHOICE QUESTIONS (20 MARKS)

(INSTRUCTION: Please answer all 20 questions in the answer sheet on page 6.)

1) The following *term* is also known as Monitor mode _____, **EXCEPT:**

A) supervisor mode
B) system mode
C) user mode
D) kernel mode

2) Data, physical machine, electronic component and software are integrated in _____.

A) computer system
B) computer operating system
C) computer architecture
D) computer organization

3) Which of the following instruction should be privileged?

A) Read the clock.
B) Issue a trap instruction.
C) Switch from user to monitor mode.
D) Access I/O device.

4) All the following statements are TRUE about Virtual Machines (VM) EXCEPT:

A) VM is a software implementation of a machine that executes programs like a physical machine.
B) VM provides hardware interface to processes whereby each process is provided with a copy of virtual host computer (hardware).
C) Computer physical resources are shared to provide VM.
D) Provides partial protection for system resources because each VM is separated from another VM.

5) Each process has a Process Control Block (PCB). The PCB will _____.

A) deletes information related to a process.
B) stores information related to a process.
C) duplicates information related to a process.
D) makes a process jump to other new process.

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The following questions 6 and 7 are referring to Figure 1.

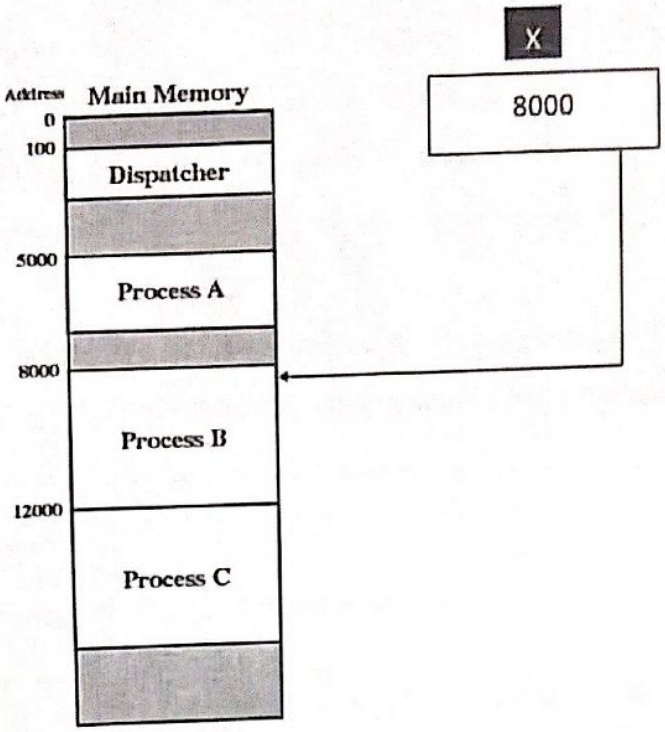


Figure 1

6) Based on Figure 1, the register labeled as **X** holds the memory address of the next instruction to be executed by central processing unit. What is the name of the register?

A) Status Register.
B) Accumulator.
C) Memory Address Register.
D) Program Counter.

7) What is the function of Dispatcher in Figure 1?

A) It is a program that invokes the processor from sleep mode.
B) It is a program that makes a process to halt.
C) It is a program that switches the processor from one process to another.
D) It is a program that sends several flags information among registers.

8) Y will initialize the process control block and allocate space for the process. What is Y?

A) Process Termination.
B) Process Switching.
C) Process Creation.
D) Process Interrupt.

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9) Which of the following statement is **FALSE**, whenever a system call `fork()` is called _____.

A) A new process is created.

B) A child process has a copy of the code as parent process.

C) A child process replaces the code of the parent process.

D) A child process will do one task while parent process does another task.

10) Given a code for running `ls` command written in C language as shown in Figure 2. Which of the following statement is **FALSE** with regard to the program?

```
pid = fork();
if (pid>0){
        wait((int*)0);
        printf("ls completed\n");
        exit(0)}
if (pid == 0){
        execl("/bin/ls", "ls","-l",(char *)0);}
perror("fork failed");
exit(1);
```

Figure 2

A) `ls completed` will always be output first.

B) If path to command does not exist, `forked failed` is output.

C) The `wait()` function causes the parent to suspend its execution until the child completed.

D) `(char *)0` a null char that marks the end of the arguments.

11) Execution of the following shell command creates _____ process / processes.

```
$ ls -l | more
```

A) zero

B) one

C) two

D) three

12) Threads in one process share the following items, **EXCEPT**:

A) Memory.

B) Thread control block.

C) Code section.

D) Data section.

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13) Which of the following application requires the least number of threads?

A) Web server.

B) Web browser.

C) Date and time utility in Windows system.

D) Calculator utility in Windows system.

14) All the following are components of a process, **EXCEPT**:

A) Program data such as stack and global variables.

B) Contents of CPU registers.

C) Interrupt vector.

D) Program instruction code.

15) Which of the followings statements are **TRUE** about concurrent processes?

i. Concurrent processing only applicable in multiprocessor computer systems.

ii. Concurrent processing requires processes to be synchronized.

iii. Concurrent processing allows modularity of processing.

iv. Concurrent processes can be independent processes or cooperating processes.

A) i, ii and iii

B) ii, iii and iv

C) i, iii and iv

D) i, ii and iv

16) Which of the followings statement is **FALSE** about context switching?

A) Context switching refers to process of switching between two processes that occupy the CPU.

B) Context switching occurs, every time but not only, when a process opens a file.

C) PCBs of the two processes involved will be created during the first context switching occurs.

D) Imbalance context switching frequency increases overhead to the system.

17) Listed below are some commonly used scheduling criteria **EXCEPT**:

A) CPU Utilization for a single-user system.

B) Throughput maximization.

C) Waiting time minimization.

D) Fairness: the degree to which all the processes are given equal opportunity to execute.

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18) Which of the following statement is **FALSE** regarding non-preemptive scheduling?

A) Suitable for batch processing systems.

B) The process could be removed from CPU resulting from an interrupt.

C) The process retained control of the CPU until the process blocked or terminated.

D) The process was adequate for CPU bound rather than I/O bound processes.

19) If the time quantum of a Round Robin scheduling algorithm is set very large, the scheduling algorithm works the same as _____.

A) Priority.

B) First Come First Serve (FCFS).

C) Shortest Remaining Time (SRT).

D) Multilevel Feedback Queue (MFQ).

20) Which of the scheduling algorithm could result an aging?

i) Priority.

ii) Shortest Job First (SJF).

iii) Shortest Remaining Time (SRT).

iv) Multilevel Feedback Queue (MFQ).

A) i and ii only.

B) ii only.

C) ii and iii only.

D) iii and iv only.

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ANSWER SHEET FOR SECTION A

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MATRIC NO.:	A17CS0100	SECTION :	08

MARKS 18

Example: =A=  =C= =D=

1) =A= =B=  =D=

2)  =B= =C= =D=

3) =A= =B= =C= 

4) =A=  =C= =D=

5) =A=  =C= =D=

6) =A= =B= =C= 

7) =A= =B=  =D=

8) =A= =B=  =D=

9) =A= =B=  =D=

10)  =B= =C= =D=

11) =A= =B=  =D=

12) =A=  =C= =D=

13) =A= =B=  =D=

14) =A= =B=  =D=

15) =A=  =C= =D=

16) =A= =B=  =D=

17)  =B= =C= =D=

18) =A=  =C= =D=

19) =A=  =C= =D=

20)  =B= =C= =D=

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SECTION B

STRUCTURED QUESTIONS (80 MARKS)

(INSTRUCTION: Please answer all 5 questions in the this question booklet.)

QUESTION 1 (10 Marks)

(a) What are the three main purposes of an operating system? [3 marks]

acts as a resource manager ✓
acts as a control program ✓
an extended machine ✓

2✓

(b) Describe TWO (2) differences between operating systems for mainframe computers and personal computers? [4 marks]

Mainframe Computer	Personal Computer
large machine	small machine
application limited to large computer cente-	application is wide

1

1

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(c) Discuss the main difficulty that a programmer must overcome in writing an operating system for a real-time environment? [3 marks]

- reliability is key
- fast and time limit sensitive
- must be 100% responsive, 100% of the time
- use in time-critical environment

QUESTION 2 (20 Marks)

(a) Answer the following questions according to Figure 3.

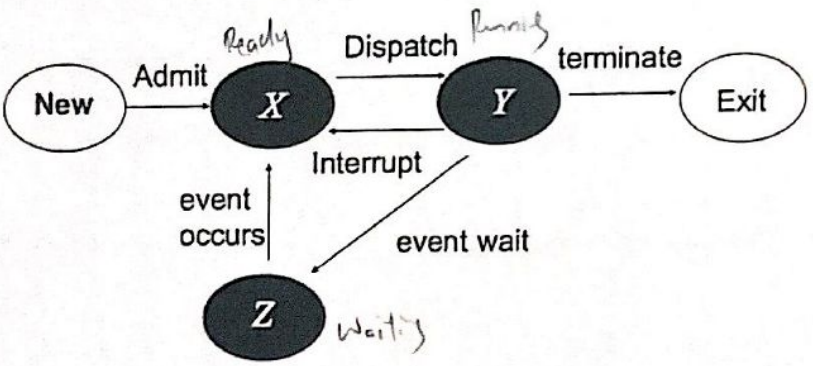


Figure 3: Five-state model of process.

i) Complete the following Table 1. [3 marks]

Table 1

Labels	State
X	Ready
Y	Running
Z	Waiting

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ii) What will happen to a process in state **Y**? In which computer resource the process resides? [2 mark]

Instructions are being executed

Process execution in CPU

iii) Give an example of an event that causes process to change its state from **Y** to **Z**? [1 mark]

I/O interrupt

✓ I/O request

I/O completion

(b) In order to keep track of events occur in computer system, operating system has to keep the information in FOUR tables. List all the tables. [2 marks]

I/O table

File table

memory table

Process table

(c) What is the different between Interrupt and Trap? [2 marks]

Interrupt is a process that terminate for awhile in between.

Trap

Interrupt - ~~total~~ eventual to currently running process

Trap - relate to error generated within currently running process

(d) Gives three (3) examples of events that may give control to the operating system or in other words, events that cause process change? [3 marks]

time slide expired

fork at child

wait for an integrated

I/O request

clock interrupt

I/O interrupt

trap

memory fault

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(e) Figure 4 illustrates a process states for trace 3 processes, A, B and C that running concurrently up to 68 clock cycles. The addresses of these processes and the dispatcher in memory are shown in Table 2.

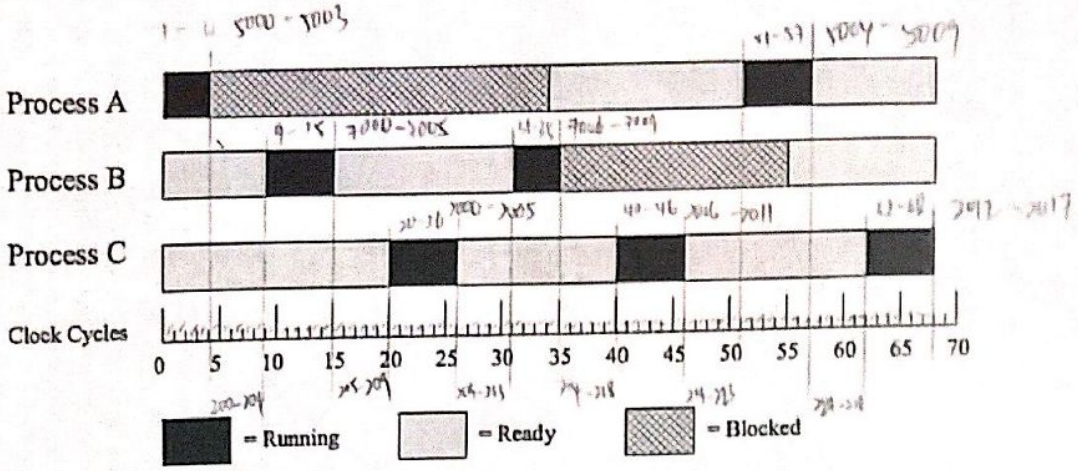


Figure 4: The process state for tracing.

Table 2

Process	Address
A	5000 - 5020
B	7000 - 7050
C	2000 - 2015
Dispatcher	200 - 204

Complete the Table 3 by writing the address of instruction that will be executed on a given clock cycle number. Assume that this is a single processor system, and maximum time quantum of 6 clock cycles. [7 marks]

Table 3

Clock Cycle	Address
1	5000
2	5001
3	5002
4	5003
5	200
6	201
7	202
8	203
9	204
10	7000
.....	.....
35	214
36	215

Clock Cycle	Address
37	216
38	217
39	218
40	2006
41	2007
42	2008
43	2009
44	2010
.....	.....
50	223
51	5004
52	5005
53	5006

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QUESTION 3 (11 Marks)

The Question 3(a) and 3(b) are referring to Figure 3 of five-state model process in Question 2 on page 8.

(a) Process Manager in operating system may be divided into two sub-managers, which are Job Scheduler (JS) and Process Scheduler (PS). Complete the Table 4 by writing the responsible sub-manager for each statement below. [5 marks]

Table 4

Statements	Sub-Manager
Responsible to schedule the processes	PS ✓
Select incoming job from Job Queue and place it into Process Queue.	JS ✓
Determine when and how long CPU is allocated to a job.	PS ✓
One of its goals is to achieve efficient system resource utilization.	JS ✓
Determine job termination	PS ✓

4

(b) Process scheduling decision may take place when a process changes from one state to another state. Write TWO of transition states that make the process scheduling preemptive. [2 marks]

Job state RUNNING to READY

Process state ✗ WAITING to READY

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(c) Process execution consists of two cycles or burst. Given a coding of C program for adding *n* numbers as shown in Figure 5, complete the figure 5 by writing the name of related cycle in the space provided. [4 marks]

```
#include <stdio.h>

int main()
{
   int n, sum = 0, c, value;

   printf("Enter the number of integers \n");
   scanf("%d", &n);

   printf("Enter %d integers\n",n);

   for (c = 1; c <= n; c++)
   {
      scanf("%d",&value);
      sum = sum + value;
   }

   printf("Sum of entered integers = %d\n",sum);

   return 0;
}
```

i) CPU burst

ii) I/O burst

iii) I/O burst and CPU burst

4

Figure 5

9

QUESTION 4 (15 Marks)

(a) Referring to program given in Figure 6, determine the output and draw the process tree (process identifier and the output, assumed process identifier begin with integer 21). [10 marks]

```
int main(int argc, char *argv[])
{
   int p1, p2, p3,p4;
   printf(" Good luck\n");
   p1 = fork();
   p2 = fork();
   p3 = fork();
   printf(" Good day\n");
}
```

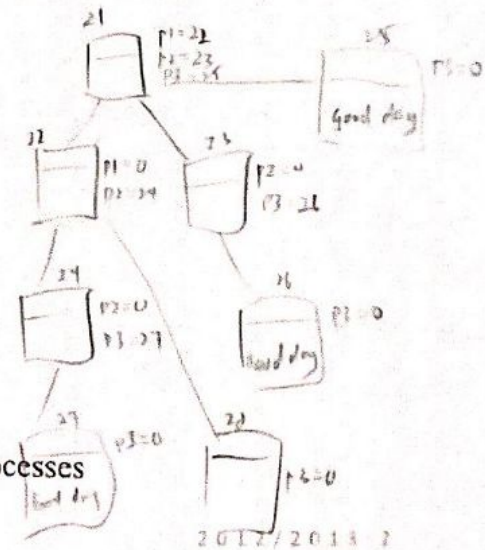
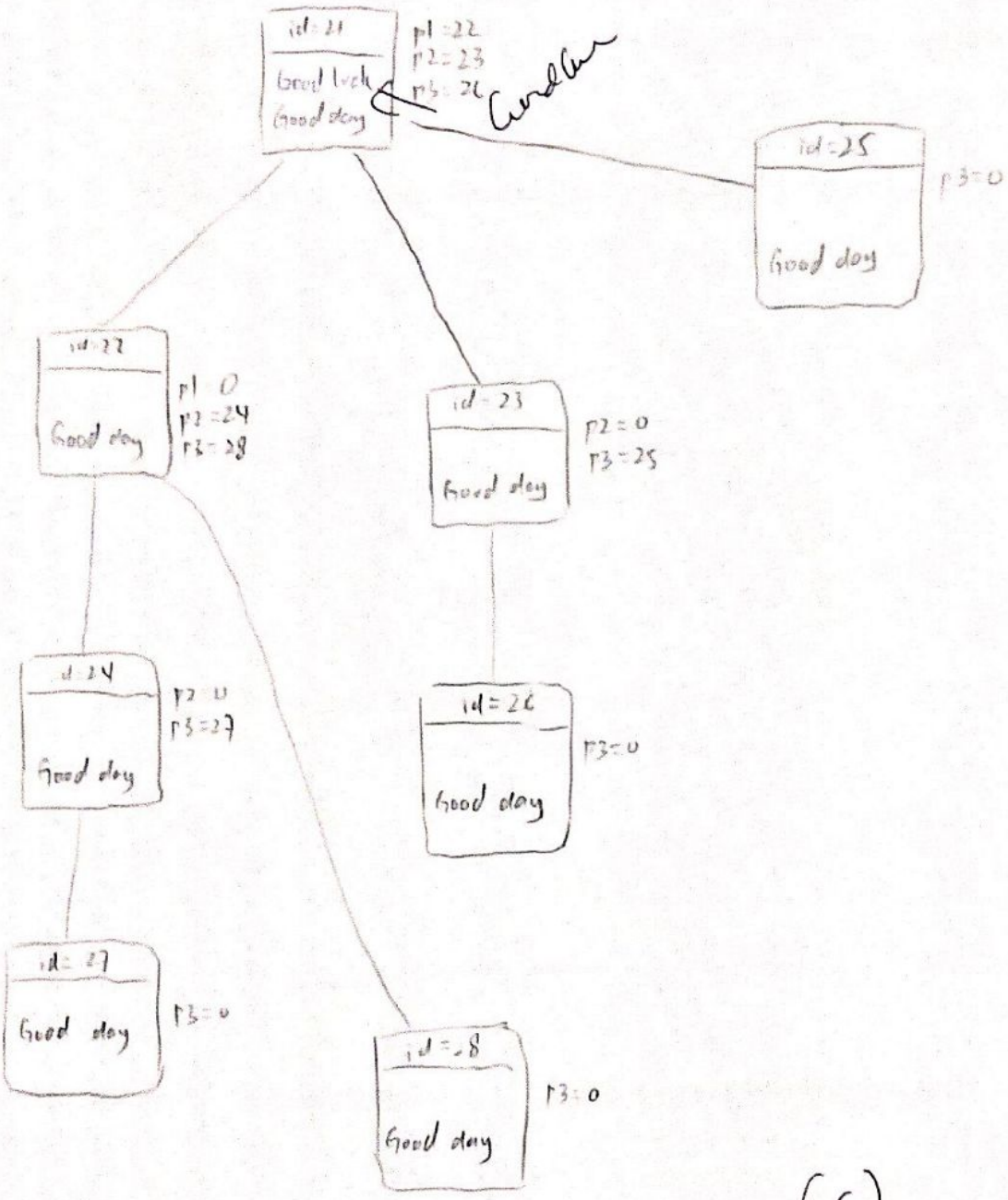


Figure 6: Code segment for creating Unix processes

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This page is purposely blanked for answering Question 4(a)



Output :

Good luck
Good day
Good day
Good day
Good day
Good day
Good day
Good day
Good day

x → 8

- process (6)
- o/p (3.5)

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